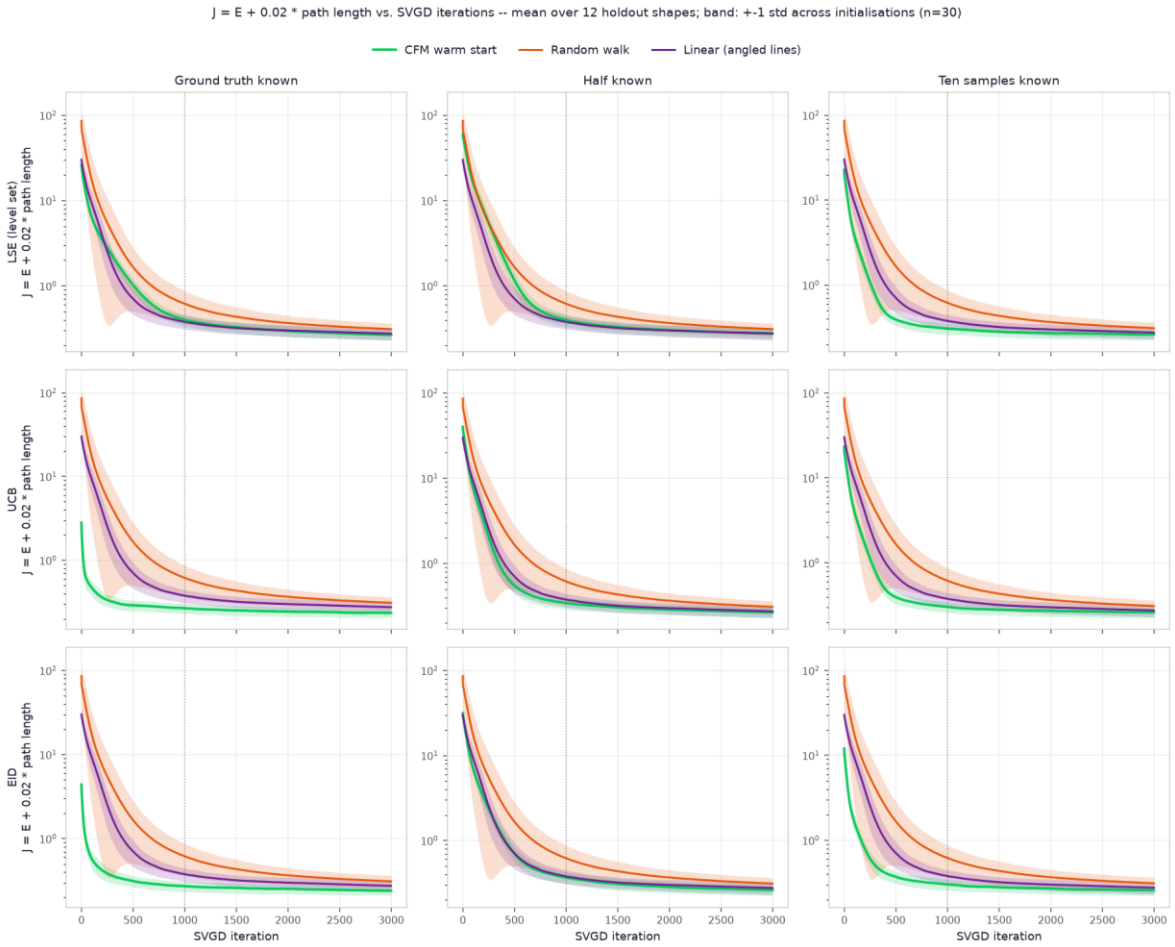
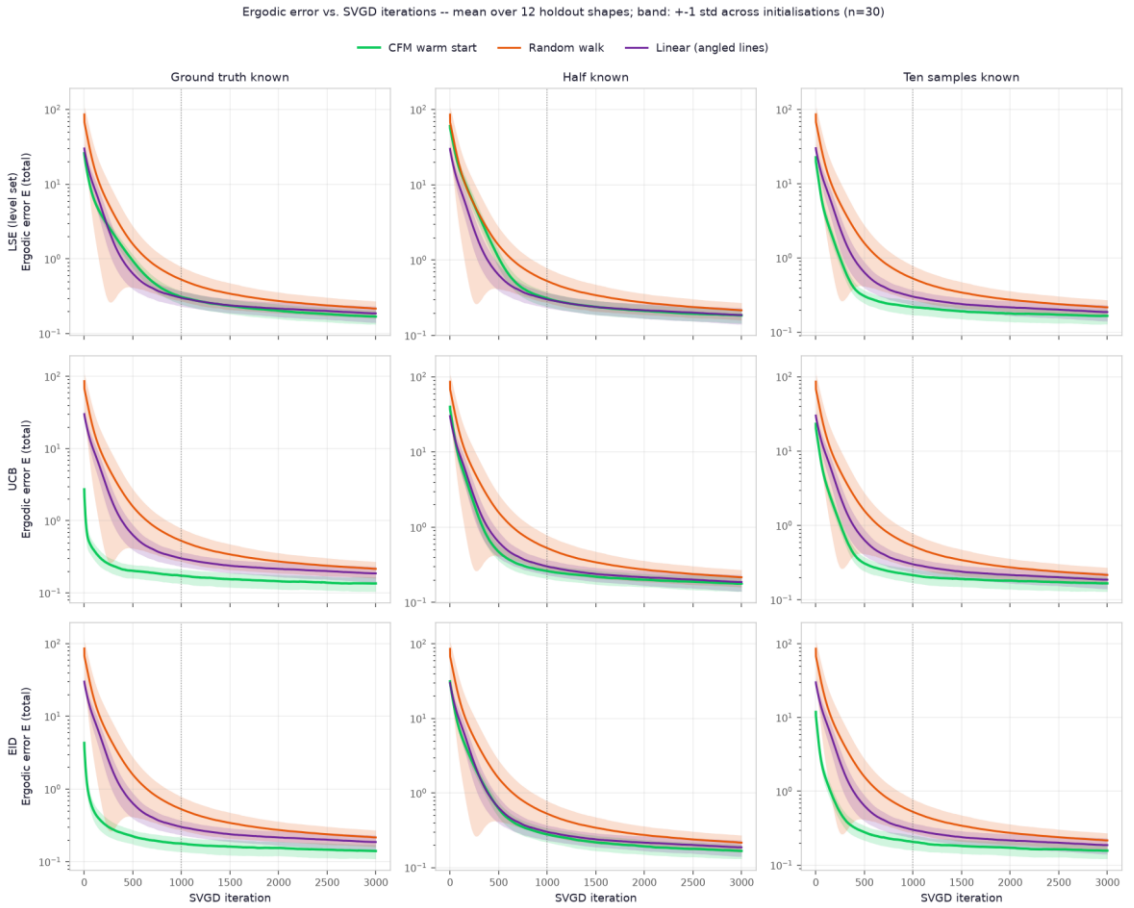
Meeting notes 11.21

Things done

- Extended training DB:
  - Instead of only training on target distributions, randomly create unknown regions on workspace with the different target distribution
  - target distribution + unknown region represented as GP
  - Create a teacher trajectory on GP
  - Training finished this morning, no testing done with new model
- More evaluation
- Comparison to Flow Matching Ergodic Coverage

Evaluation svdg iteration untill ergodicity / J convergence

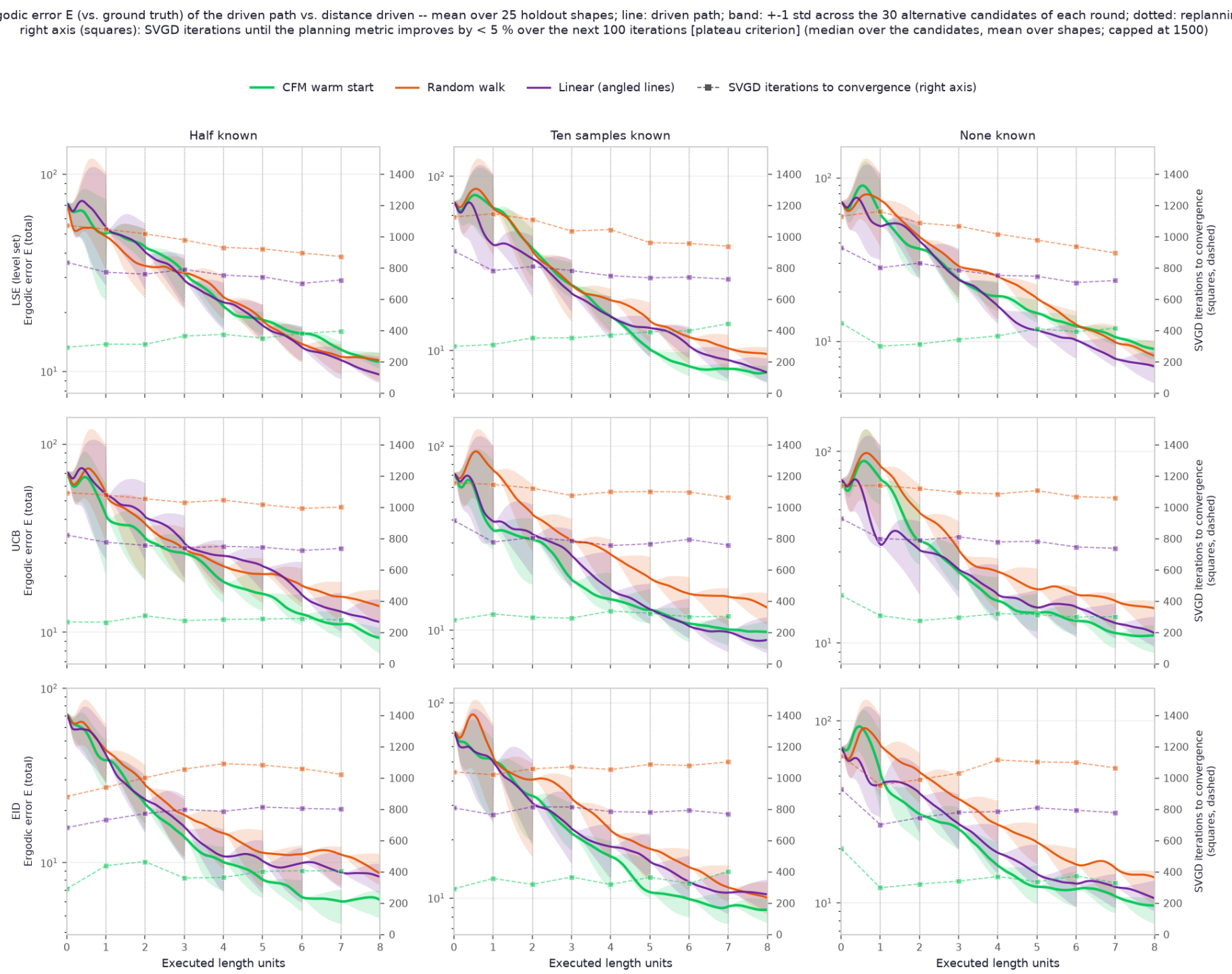




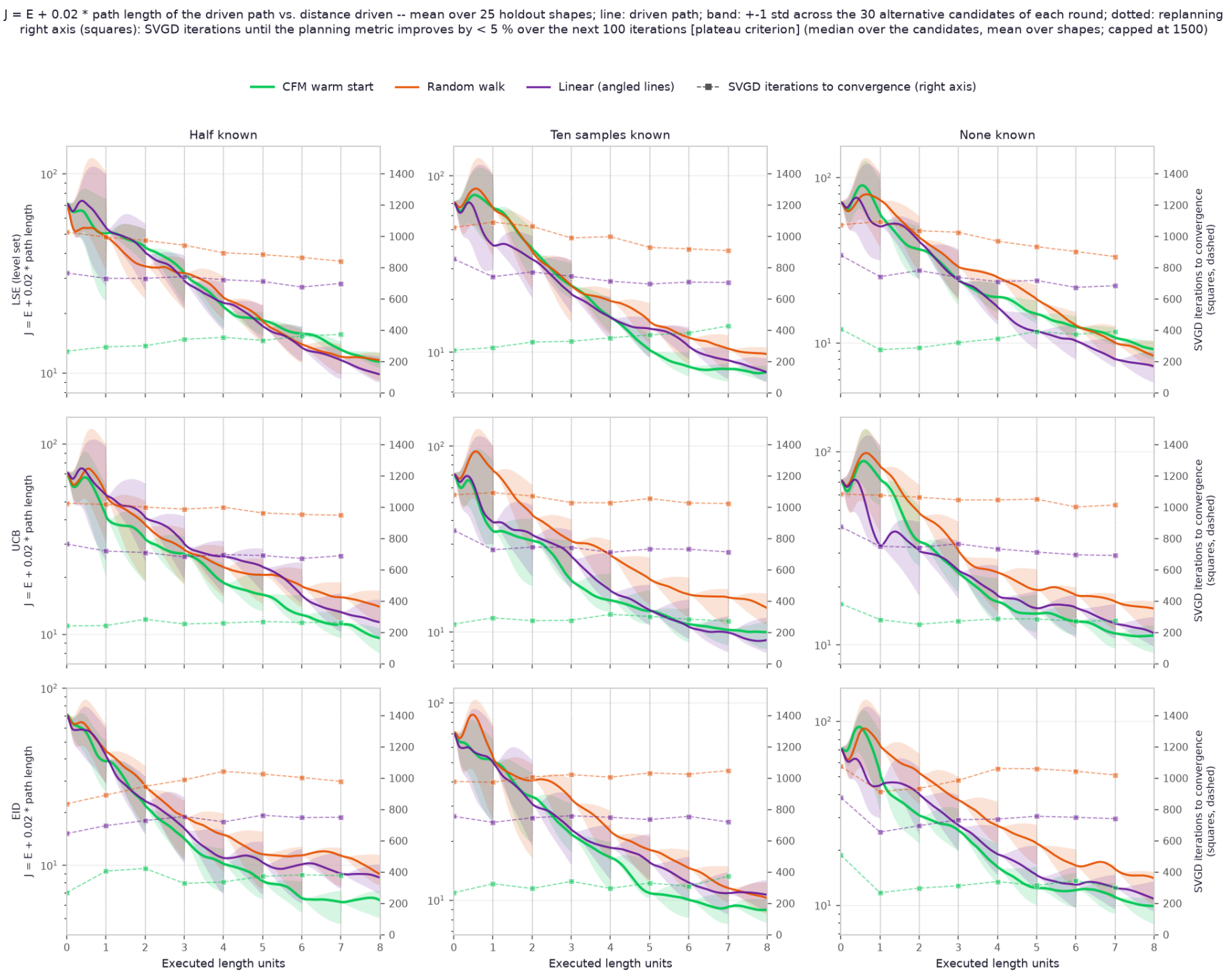
Evaluation

Replanning system:

- Execute length unit, update, replan
- Developement of ergodicity



Evaluation


J = E + 0.02 * path length of the driven path vs. distance driven -- mean over 25 holdout shapes; line: driven path; band: +-1 std across the 30 alternative candidates of each round; dotted: replanning
right axis (squares): SVGD iterations until the planning metric improves by < 5 % over the next 100 iterations [plateau criterion] (median over the candidates, mean over shapes; capped at 1500)
CFM warm start
Random walk
Linear (angled lines)
SVGD iterations to convergence (right axis)
Half known
Ten samples known
None known
LSE (level set)
J = E + 0.02 * path length
UCB
J = E + 0.02 * path length
EID
J = E + 0.02 * path length
SVGD iterations to convergence (squares, dashed)
Executed length units

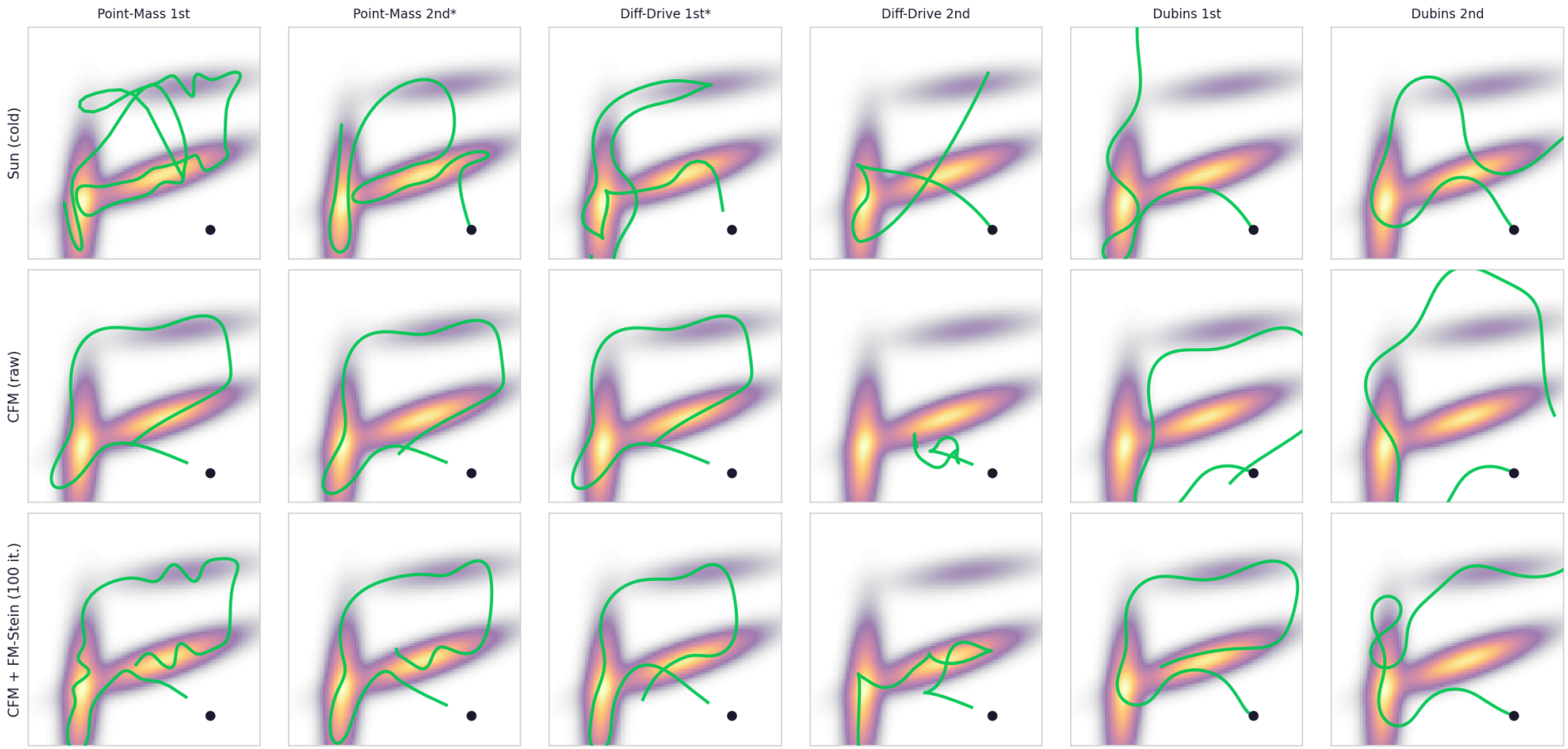
Target shape recovery



Recreate evaluations from sun et al

- Problems to recreate/understand fig 6
- Very short explaination, no code comparison
- There is no way it could know about the target distribution in the lower center

Comparison to fig 11



Researchquestions

- Does the CFM intialization allow for a faster **ergodicity convergencs in general**? – Yes
- Does the CFM intialization allow for a faster **ergodic coverage of unknown regions**? - Yes
- Does the CFM intialization allow for a faster recovery of unknown regions? – Not really – Yet?

Plans for next week

- Test new model
- Implement MuJoCo simulation for a realistic lab setting
- Open Questions: How to deal with 3D obstacles for the robot arm, not only the endeffector?